Efficient Swarm Intelligence Algorithm for Optimizing Multi-Stage Designs for Phase II Clinical Trials

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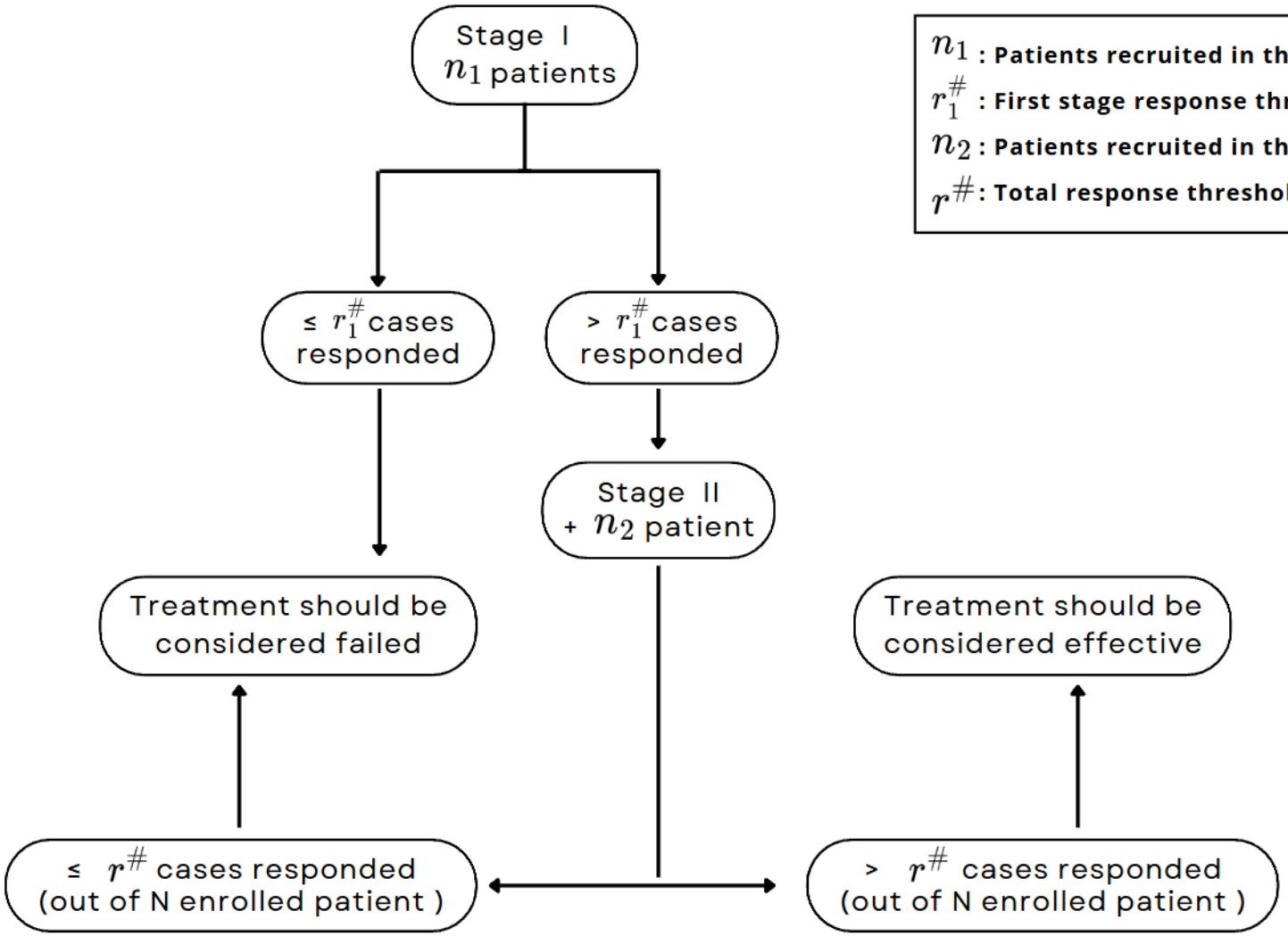
November 19, 2024

Research Background

- *Importance of Multi-Stage Design*:
  - Enhances cost-efficiency, resource savings, *and* trial effectiveness.
- *Simon's Two-Stage Design*:
  - Enables early termination, saving resources and reducing patient risk (Simon 1989).
- *Why Three-Stage Design?*:
  - *Flexible Evaluation*: Allows early stopping (Chen 1997).
  - *Optimized Sample Use*: Closer to ideal allocation.
  - *Lower Error Risk*: Reduces false positives/ negatives.

Literature Review: Simon's Two-Stage Design



n_1 : Patients recruited in the first stage
$r_1^{\#}$: First stage response threshold
n_2 : Patients recruited in the second stage
$r^{\#}$: Total response threshold

Literature Review: Simon's Two-Stage Design

Phase II Trials Based on Hypothesis Testing

- $H_0 : p \leq p_0$
- $H_1 : p \geq p_1$

- p_0: Disinterested response level
- p_1: Desired response level

Considerations

- Constrained by α and β
- Minimizing the number of patients receiving low-efficacy treatment

Probability of Early Termination at Each Stage

- Assume each patient's response is a binary random variable with a success probability of p.
- Let $b(x; n, p)$ denote the probability of having x responses among n patients. The binomial distribution formula is:

$$b(x; n, p) = \binom{n}{x} p^x (1-p)^{n-x}$$

Probability of Early Termination (PET) in K-Stage Design

- The formula is as follows:

$$PET_k(p) = \underbrace{\sum_{x_1=L_1}^{U_1} \sum_{x_2=L_2}^{U_2} \cdots \sum_{x_{k-1}=L_{k-1}}^{U_{k-1}}}_{k-1 \text{ Summations}} \left[\prod_{s=1}^{k-1} b(x_s; n_s, p) \right] B\left(r_k = \sum_{s=1}^{k-1} x_s, n_k, p \right)$$

- where:
  - $b(x_s; n_s, p)$ represents the binomial probability at stage s.
  - L_j and U_j are the lower and upper bounds of stage j, respectively.
  - $B(\cdot)$ is the cumulative distribution function of the binomial distribution.

Probability of Treatment Rejection and Expected Sample Size

- **Probability of Treatment Rejection at the End of the Third Stage**:

$$\sum_{x_1=r_1+1}^{\min[n_1,r_3]} \sum_{x_2=r_2+1-x_1}^{\min[n_2,r_3-x_1]} b(x_1;n_1,p)b(x_2;n_2,p)B(r_3-x_1-x_2;n_3,p)$$

- **Expected Sample Size (EN)**:

$$EN = n_1 + (1-PET_1)n_2 + (1-PET_2)n_3$$

Total Combination Possibilities

- 2-Stage Design: $1,105,524$ possibilities
- 3-Stage Design: $213,672,935$ possibilities
- 4-Stage Design: $22,546,940,047$ possibilities
- 5-Stage Design: $1,530,973,869,444$ possibilities

As the number of stages increases, the number of combinations becomes larger, making clinical design more complex.

Note: Enumeration range:
Total sample size $N = n_1 + \ldots + n_k$, where $30 \leq N \leq 70$.

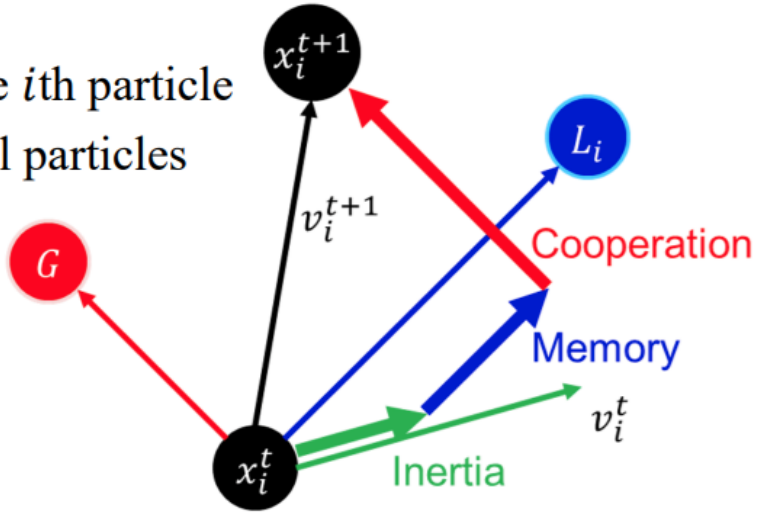
PSO(Particle Swarm Optimization)

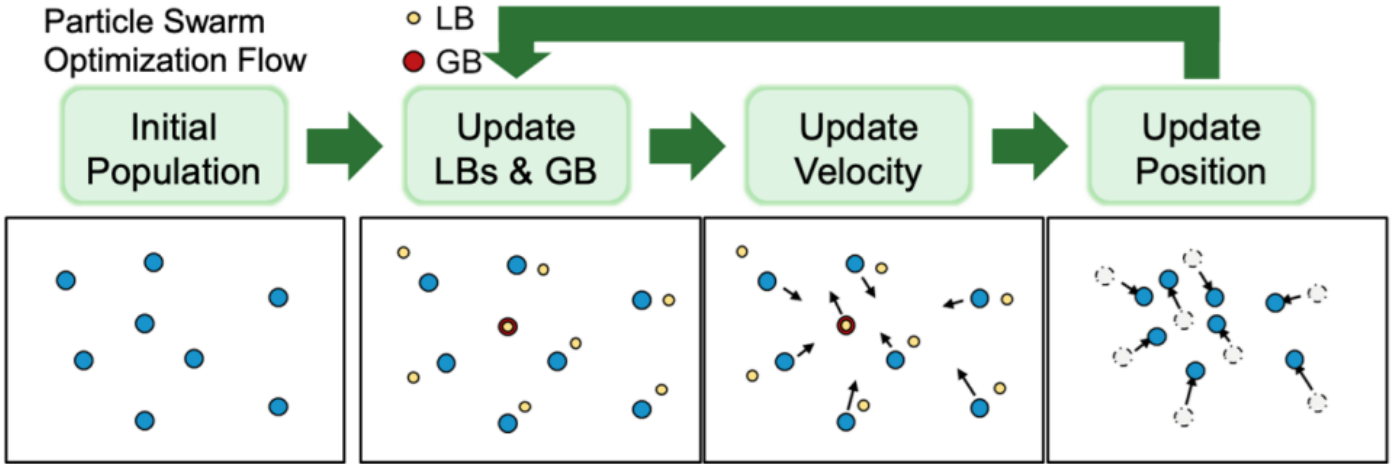
- Local Best L_i: historical best solution of the ith particle
- Global Best G: historical best solution of all particles

$$v_i^{t+1} = w \cdot v_i^t + c_1 \cdot r_1(L_i - x_i^t) + c_2 \cdot r_2(G - x_i^t)$$

$$x_i^{t+1} = x_i^t + v_i^{t+1}$$

- w, c_1, c_2 are tuning parameters.
- r_1, r_2 are uniform random vectors.





Four-stage Optimal Design

p_0	p_1	Stage 1	Stage 2	Stage 3	Stage 4	EN(p_0)	$PET_1(p_0)$	$PET_2(p_0)$	$PET_3(p_0)$
0.05	0.25	0/9	0/17	1/18	2/26	13.78	0.63	0.63	0.82
0.1	0.3	0/11	1/15	2/19	4/26	17.37	0.31	0.57	0.73
0.15	0.35	0/9	2/16	4/26	7/33	20.55	0.23	0.58	0.72
0.2	0.4	1/10	4/21	7/30	11/43	22.73	0.38	0.63	0.8
0.25	0.45	1/10	4/18	8/28	14/45	24.42	0.24	0.54	0.78
0.3	0.5	2/11	6/21	11/33	17/46	25.77	0.31	0.59	0.78
0.35	0.55	4/14	10/27	15/38	20/48	26.55	0.42	0.7	0.82
0.4	0.6	4/13	10/24	16/36	23/49	26.29	0.3	0.62	0.75
0.45	0.65	3/10	9/20	16/33	25/48	25.81	0.27	0.61	0.77
0.5	0.7	4/11	8/17	15/28	26/45	24.96	0.27	0.52	0.75
0.55	0.75	5/11	10/18	19/32	28/45	23.2	0.37	0.63	0.76
0.6	0.8	5/10	9/15	17/27	27/40	20.91	0.37	0.61	0.76
0.65	0.85	5/9	13/19	20/28	24/33	18.32	0.39	0.72	0.75
0.7	0.9	1/3	6/9	15/20	24/31	14.72	0.22	0.56	0.8
0.75	0.95	3/5	6/8	11/14	16/19	10.23	0.37	0.63	0.77
$p_1 - p_0 = 0.20$ and $\alpha = 0.1$, $\beta = 0.1$									

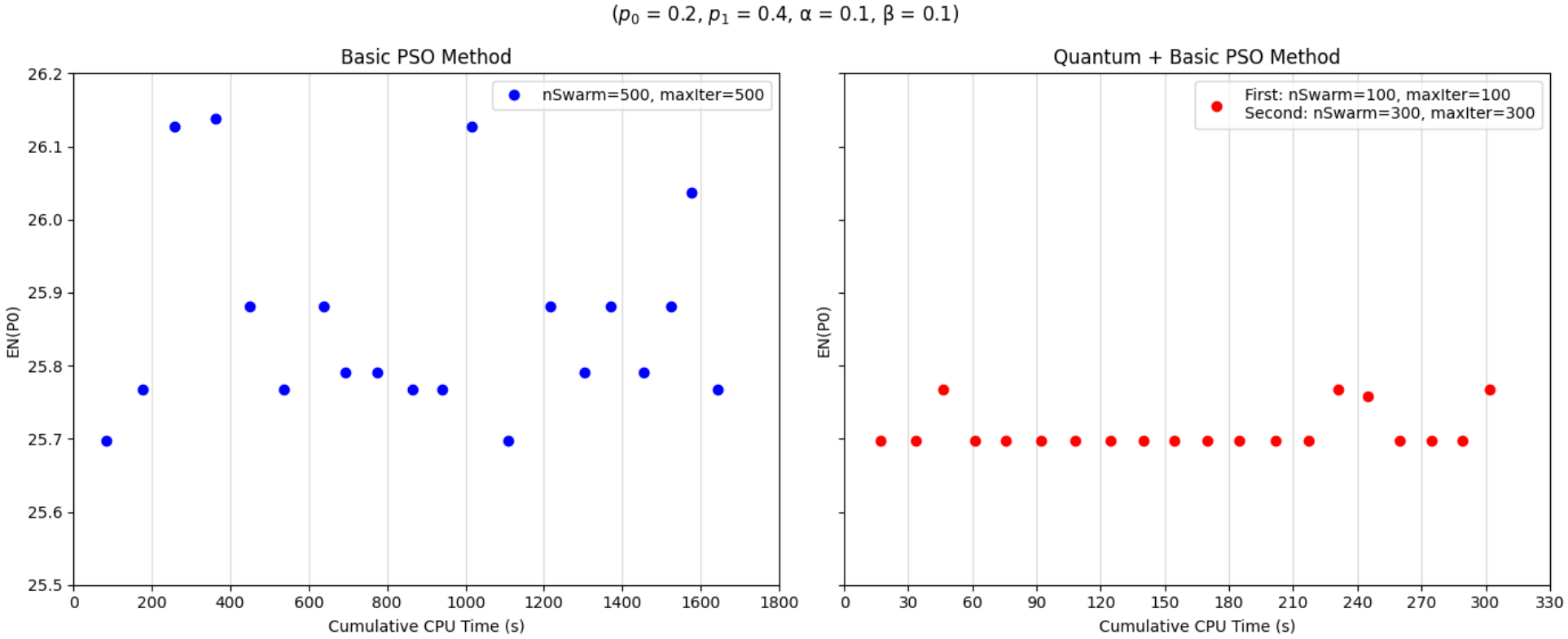
Four-stage Minimax Design

P_0	P_1	Stage 1	Stage 2	Stage 3	Stage 4	EN(p_0)	$PET_1(p_0)$	$PET_2(p_0)$	$PET_3(p_0)$
0.05	0.25	0/13	0/15	1/18	2/20	15.86	0.51	0.51	0.79
0.1	0.3	0/12	1/16	3/24	4/25	18.83	0.28	0.53	0.8
0.15	0.35	0/11	2/17	4/25	7/32	21.75	0.17	0.53	0.72
0.2	0.4	1/13	3/20	6/29	10/36	25.7	0.23	0.44	0.67
0.25	0.45	8/29	9/31	11/35	12/37	30.71	0.71	0.78	0.87
0.3	0.5	3/20	8/30	12/36	15/39	33.1	0.11	0.44	0.74
0.35	0.55	3/14	6/20	10/27	18/42	27.36	0.22	0.43	0.68
0.4	0.6	6/20	9/25	12/30	20/41	30.82	0.31	0.52	0.73
0.45	0.65	5/15	8/20	14/30	22/41	27.97	0.26	0.43	0.67
0.5	0.7	5/14	10/22	17/32	23/39	27.93	0.21	0.44	0.72
0.55	0.75	6/13	13/24	22/35	24/38	25	0.36	0.59	0.87
0.6	0.8	4/9	12/20	16/25	24/35	21.54	0.27	0.61	0.75
0.65	0.85	7/12	11/17	15/22	23/31	19.32	0.42	0.6	0.73
0.7	0.9	5/9	11/16	15/20	20/25	17.02	0.27	0.56	0.77
0.75	0.95	3/5	6/8	11/14	16/19	10.23	0.37	0.63	0.77
$p_1 - p_0 = 0.20$ and $\alpha = 0.1$, $\beta = 0.1$									

EN(p_0) Comparison between 3 & 4-stage Designs

P_0	P_1	EN(p_0) (Optimal)		Variation	EN(p_0) (Minimax)		Variation
		3-stage	4-stage		3-stage	4-stage	
0.05	0.25	13.78	13.78	0	15.86	15.86	0
0.1	0.3	17.79	17.37	-0.42	19.1	18.83	-0.27
0.15	0.35	21.11	20.55	-0.56	22.23	21.75	-0.48
0.2	0.4	23.86	22.73	-1.13	26.41	25.7	-0.71
0.25	0.45	25.55	24.42	-1.13	29.83	29.1	-0.73
0.3	0.5	26.75	25.77	-0.98	33.71	33.1	-0.61
0.35	0.55	27.84	26.55	-1.29	28.59	27.36	-1.23
0.4	0.6	29.81	26.29	-3.52	32.48	30.82	-1.66
0.45	0.65	27.22	25.81	-1.41	28.94	27.97	-0.97
0.5	0.7	26.37	24.96	-1.41	29	27.93	-1.07
0.55	0.75	24.36	23.2	-1.16	27.74	25	-2.74
0.6	0.8	22.3	20.91	-1.39	22.97	21.54	-1.43
0.65	0.85	19.23	18.32	-0.91	22.22	19.32	-2.9
0.7	0.9	15.45	14.72	-0.73	17.85	17.02	-0.83
0.75	0.95	10.94	10.23	-0.71	10.94	10.23	0

EN(p_0) vs Cumulative CPU Time for Different PSO



Discussions, Challenges, and Future Plans

- **Challenges in Algorithm Implementation**:
  - Stability issues, convergence speed, and sensitivity to parameters .
  - Inconsistent results across multiple runs.
- **Future Research Directions**:
  - Enhancing algorithm stability and convergence efficiency.
  - Investigating other relevant clinical trial designs or incorporating additional literature for further improvements.

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